Superintelligent Agent: The Foundational Unit of Production and Life in Future Human Society

Jianjun Wu[1], Xiao Liu[2], Merouane Debbah[3], Kwok Po NG[4], Li Sun[5], Pietro Lio[6], Yu Guang Wang[2,7]

1.Asmote; 2.Shanghai Jiao Tong University; 3.Khalifa University; 4.Hong Kong Baptist University; 5.Peng Cheng Laboratory; 6.University of Cambridge; 7.Toursun Synbio

Abstract

The Superintelligent Agent (SIA) is envisioned as the foundational unit of future human society, integrating perception, cognition, decision-making, and execution beyond human capability. Emerging from the convergence of AI for Science and AI for Industry, it fuses superhuman cognition, embodied interaction, and collective coordination into an autonomous cyber-physical organism. Its architecture mirrors biological systems— Central Control Unit (brain) governs reasoning, Specific agents (Cerebellums) handle specialized tasks, the Nervous System (5G/6G) ensures ultra-low-latency connectivity, Terminals (robots and sensors) act in the physical world, and the Knowledge Engine sustains continuous learning. The AI Edge infrastructure is a typical example of superintelligent agents, which enables real-time intelligence by merging communication, sensing, and computation into a distributed multi-agent system capable of self-learning and adaptation. Two major realizations exemplify this paradigm: the Scientific Superintelligent Agent, which integrates foundation models with automated wet labs to accelerate discovery, and the Industrial Superintelligent Agent, which reshapes production through autonomous collaboration and intelligent manufacturing. Together, they redefine the relationship between humans, machines, and knowledge, marking a transition from data-driven to physics-informed intelligence and from centralized control to collective autonomy. With responsible

governance, the SIA promises to inaugurate a new era of automated discovery, sustainable industry, and amplified human creativity.

1. Introduction

As artificial intelligence technology undergoes leapfrog development, we stand at the cusp of a singularity comparable to the Industrial Revolution, redefining the boundaries of dreams and reality at an ever-accelerating pace [60]. At the forefront of scientific exploration, AI for Science is instigating a paradigm shift [59]. Predictive models, exemplified by DeepMind's AlphaFold, have not only solved the protein folding problem that has perplexed the biological community for half a century with atomic-level accuracy, but have also accelerated the speed of new drug discovery and disease mechanism research by several orders of magnitude. Its profound impact led to its recognition with the 2024 Nobel Prize in Chemistry. In the arteries of the real economy, AI for Industry is quietly accumulating its strength. Advanced robotics, with Embodied AI at their core, are moving beyond enclosed production lines and into the complex real world, heralding a new paradigm of productivity where the nature of labor is completely subverted. These advancements are not isolated technological innovations; they are powerful technological waves converging from different dimensions, collectively pointing toward a profound, intelligence-driven vision of the future, a new breed of artificial intelligence which can upend centuries of scientific tradition. The construction of this vision stems from a fundamental challenge that we must confront: the shifting trends in global population structure. In this new societal form, the continuation and prosperity of human civilization urgently require solutions to two core issues:

- **The Quality and Length of Life:** How can we leverage the power of technology to transcend the natural limits of lifespan and enhance the health and quality of life for all humanity?

- **The Efficiency and Purpose of Production:** How can we liberate humans from repetitive and exhaustive production processes to pursue higher levels of creation and exploration?

It could be pre-defined that scientific progress is fast and unrelenting. For many decades, progress has actually slowed [2]. In response to these ultimate questions, a clear evolutionary path is emerging. The first step is to focus on life itself through **AI for Life Science**, addressing the fundamental demand to “live longer and live better”. The second step is to delve into the front lines of production through **AI for Industry**, solving the efficiency problem of “doing more and doing it more easily". It is noteworthy that while modern communication technologies (5G/6G) have, to some extent, freed people from the "production site", they have not fundamentally liberated them from the "production process". True liberation requires a form of intelligence far more powerful than a mere tool.

The continuous development and iteration of these first two steps will ultimately give rise to a new organizational form that empowers all industries—the **Superintelligent Agent**. This is not a single algorithm or piece of hardware, but a complex intelligent system integrating perception, cognition, decision-making, and execution capabilities, serving as the fundamental unit of production and life in future human society. Its mission is to provide **AI for All** services, fundamentally reshaping industrial structures and lifestyles, and ultimately achieving a holistic elevation of human well-being.

The importance of Artificial Intelligence (AI) in contemporary society is increasingly prominent, with its influence permeating scientific discovery and research, industrial manufacturing and production, and various vertical domains [3,4]. As AI technology continues to evolve, we are progressively advancing toward more general and comprehensively capable forms of advanced intelligence [5]. The Superintelligent Agent, as a higher-order form on this evolutionary path, heralds a future where AI systems surpassing human intellect drive industrial production and scientific discovery. Although the fully realized Superintelligent Agent currently remains in the conceptual stage, its potential for disruptive impact has garnered widespread attention from both academia and industry.

We aim to provide an in-depth exploration of the core concepts of Scientific Superintelligent Agents and Industrial Superintelligent Agents. We will begin with their definitions and key characteristics, analyzing their revolutionary applications in automated scientific research and intelligent industrial production. Special emphasis will be placed on establishing a closed-loop interaction between Scientific Superintelligent Agents and automated wet-lab platforms, as well as advancing edge intelligence and curated knowledge repositories to support Industrial Superintelligent Agents. By synthesizing existing theories and practices, this paper endeavors to delineate a clear and actionable roadmap for the development of Superintelligent Agents, ultimately defining a platform that can serve both science and industry.

2. What is Superintelligent Agent?

The Superintelligent Agent (SIA) can be defined as an autonomous, cognitive–physical system that not only surpasses human intelligence but also interacts dynamically with its environment to accomplish complex goals. Its distinguishing traits can be summarized in three interdependent dimensions:

Superior Intelligence Beyond Human Capability. The SIA exhibits cognitive performance that exceeds individual human capacity in one or multiple dimensions—reasoning, learning, problem-solving, or creativity. It can integrate multimodal information, formulate hypotheses, and generate novel solutions that transcend the limits of human intuition or expertise.

Interactive Coupling with the Physical World. Unlike traditional disembodied AI systems, the SIA possesses the ability to perceive, act, and adapt within physical environments. Through continuous interaction, it acquires real-world data and experiential knowledge, forming a closed perception–action loop that enables environmental adaptation, causal understanding, and long-term self-improvement.

Multi-Agent Integration and Collective Intelligence. The SIA rarely exists as a single monolithic entity; rather, it functions as a network of intelligent agents that cooperate through communication and task decomposition. By leveraging distributed perception, computation, and memory, it can autonomously gather, exchange, and synthesize information across domains—achieving emergent intelligence greater than the sum of its parts.

In essence, the Superintelligent Agent represents a convergence of superhuman cognition, embodied interaction, and collective coordination, forming the foundational architecture for the next stage of intelligent society.

2.1 Superintelligence of Superintelligent Agent

Whereas artificial intelligence (AI) is generally understood as a system capable of passing the Turing Test, and Artificial General Intelligence (AGI) as intelligence that matches human performance across domains, ASI represents a qualitative transformation. It is not a linear extension of current systems but a hypothetical form of intelligence whose intellectual scope comprehensively surpasses human capacity. Such potential has sparked debates about its societal, ethical, and security implications.

What is Superintelligence?

Oxford philosopher Nick Bostrom defines superintelligence as “an intellect much smarter than the best human brains in practically every field, including scientific creativity, general wisdom, and social skills.” [10] This definition highlights the essential distinction between superintelligence and today's "narrow AI," which, no matter how advanced in facial recognition [55,56] or chat robot [57,58], remains confined to specific tasks. Superintelligence, in contrast, implies universal superiority—from producing profound scientific discoveries to navigating subtle emotional interactions—operating at a level unattainable by human cognition.

Core Characteristics

The transformative nature of superintelligence lies both in its unprecedented scope and in the mechanisms of its emergence. Four features are central:

- **Recursive Self-Improvement**: Once an AI reaches the capacity to improve its own architecture, it can trigger an *intelligence explosion*. Each iteration enhances its ability to further optimize itself, leading to exponential acceleration and the creation of entities far beyond collective human wisdom.
- **Unbounded Knowledge Processing**: With vast memory and processing capacity, superintelligence could absorb, synthesize, and expand upon the entirety of human knowledge, learning new concepts at speeds exceeding the limits of human collectivity.
- **Extraordinary Creativity**: Beyond knowledge retention, superintelligence could generate disruptive theories, designs, and innovations. Its problem-solving would identify essential structures of challenges and devise efficient, previously inaccessible solutions.
- **Autonomy and Predictive Insight**: By modeling complex systems, superintelligence could forecast trends, autonomously execute decisions, and potentially develop advanced forms of *emotional intelligence*, enabling nuanced participation in social dynamics.

Theoretical Pathways

Although still speculative, several pathways toward superintelligence are emerging. Advances in deep learning, reinforcement learning and large lanaguage models provide foundational tools. On the othe hand, massive, high-quality datasets in various scenarios known and unknown, and unprecedented computational power act as the essential fuel. These two dimensions together will give rise to a level of intelligence that exceeds human comprehension, driving intelligent automation across diverse real-world contexts. In turn, this automation continuously refines the underlying AI models and enhances data acquisition, forming a self-reinforcing loop that elevates scenario-specific intelligence. Crucially, this process depends on sustained and dynamic interaction with the physical world.

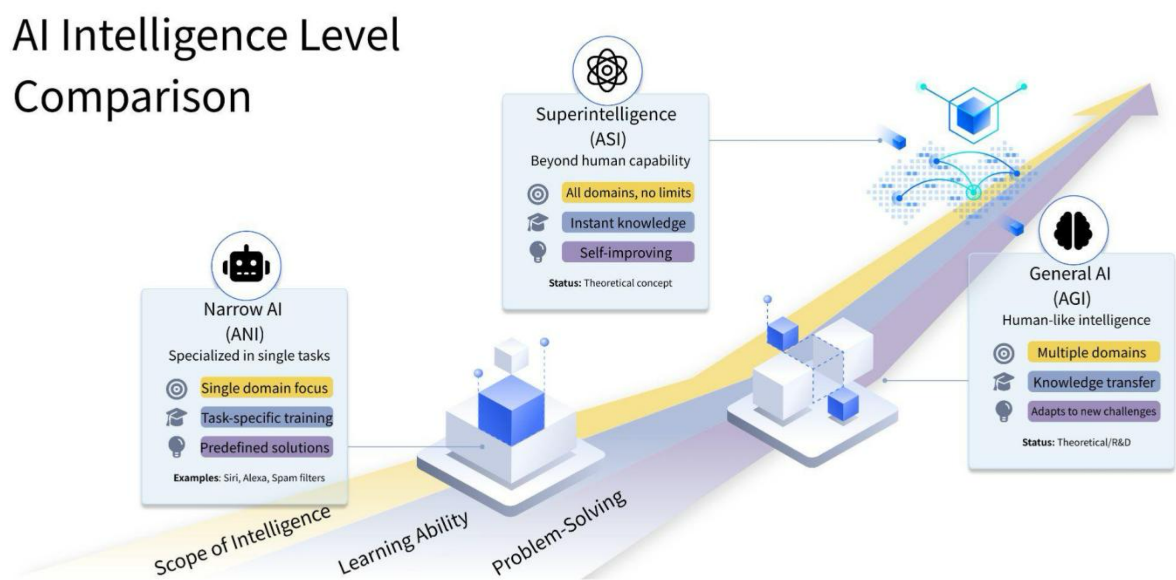

Figure 1 Comparison of Intelligence Levels

2.2 Foundation Models for Physical World Interaction

To perceive and act in the real world, SIA requires foundation models that represent, predict, and control complex, multimodal, and physics-governed environments.

World Foundation Models (WFMs) serve as digital twins, providing safe, interactive simulation to generate synthetic data, stress-test strategies, and pre-train/control policies [62]. Transformer-based diffusion models support high-fidelity scene evolution; autoregressive variants excel in sequential decision-making and policy learning.

Physics-Informed Neural Networks (PINNs) embed physical laws (often PDEs) directly into objectives or architectures, enforcing physical consistency, boosting accuracy under sparse/biased data, and reducing dependence on massive labels [63]. Control-oriented extensions translate these priors into executable policies for actuation.

These paradigms are complementary. Practical stacks hybridize them—for example: WFM for scene simulation and safety testing; STFM for non-visual signals and propagation states; and PINN-regularized objectives to preserve physical plausibility. The result is a foundation model that perceives accurately, reasons safely, and acts law-consistently.

2.3 Multi-agents of Superintelligent Agent

In artificial intelligence research, an **agent** is defined as an entity capable of perceiving its environment, making autonomous decisions, and acting toward specified goals. Multi-agent systems built upon reinforcement learning and large language models (LLMs) enable AI to tackle complex, multi-stage tasks. In this architecture, the LLM functions as the central brain, orchestrating and coordinating the actions of specialized agents. Each individual agent addresses a specific subtask, while collaborative interaction among agents ensures coherent execution across the entire problem-solving pipeline. Agent can be an AI model or a device can perceive the real scenario or chat with other intelligent device.

Building upon this foundation, the **Superintelligent Agent (SIA)** integrates the goal-directed nature of agents with the superior cognitive capabilities of superintelligence. The result is not only higher performance in intellectual tasks but also a system-level architecture with unprecedented adaptability, scalability, and collaborative power.

2.5 Architecture of Superintelligence Agent

As shown by *Figure 2*, the SIA can be deconstructed through a biological analogy, where intelligence emerges from the coordinated operation of interdependent subsystems: Central Control Unit, Domain Modules, Communication Network, Physical Interfaces, Driving Core, Knowledge and Compute Base.

Central Control Unit as Brain

It is responsible for global strategy, reasoning, and decision-making. It allocates tasks, synthesizes knowledge, and coordinates the entire system.

Domain Modules as Cerebellums

Specialized sub-centers operating in vertical domains—drug discovery, automated labs, logistics optimization—providing rapid responses through edge computing while feeding critical results back to the Brain.

Nervous System as Lifeline

The Nervous System provides the connective tissue of the SIA. Unlike traditional communication networks, it functions as an intelligent infrastructure. URLLC ensures microsecond-level responses for robots and remote surgery, eMBB supports data-intensive tasks like medical imaging, and mMTC integrates massive sensor arrays in cities or agriculture. By combining connectivity with computation, the Nervous System ensures seamless, lossless information flow between Brain, Cerebellums, and Limbs.

Limbs as Physical Terminal

The Limbs represent the SIA's embodiment in society. In manufacturing, they include inspection systems and robotic arms; in healthcare, wearable monitors and surgical robots; in cities, traffic and environmental sensors. These devices are not passive endpoints but intelligent agents capable of local preprocessing, thereby anchoring the SIA's perception–decision–execution loop in the physical world.

2.6 Embodiment of Superintelligence Agent

The Driving Core as Local Intelligence

Instead of relying exclusively on centralized computation, the Driving Core represents distributed edge intelligence that sustains real-time autonomy. It enables localized perception–decision–action loops, ensuring millisecond-level reflexes in dynamic environments. For instance, in a smart factory, a camera can detect a defect, the edge node immediately analyzes the input, and a robotic arm corrects the anomaly on the spot. This rapid loop occurs without waiting for central commands, while summarized outcomes are transmitted upward for global optimization. The Driving Core thus provides both fast local reflexes and feedback for long-term adaptability, allowing the system to balance responsiveness with strategic evolution.

The Knowledge Engine as Foundational Base

At the foundation lies the Knowledge Engine—a multi-layered ecosystem of data and computation that sustains the continuous growth of intelligence. Its role is not simply to "store fuel", but to transform unstructured streams into actionable knowledge:

- **Raw Data Streams:** unprocessed flows from sensors, devices, and scientific instruments.
- **Curated Knowledge Assets:** structured datasets and validated metadata that ensure reliability for training.
- **Optimized Intelligence Units:** distilled models, inference pipelines, and domain-specific insights that can be redeployed across edge and cloud.

Through this staged refinement, the Knowledge Engine enables continuous circulation of intelligence: raw inputs are absorbed, processed into structured knowledge, and converted into lightweight models suitable for deployment at scale. It anchors the system's evolutionary cycle of learning, distillation, and redeployment, ensuring that local reflexes and global strategies remain tightly coupled.

The Closed-Loop Paradigm

Ultimately, the SIA operates through a self-reinforcing loop of *collect* → *refine* → *empower* → *act*. Raw experience from the physical world becomes structured intelligence in the digital sphere, which is then reinjected into embodied agents. This architecture allows the SIA to continuously learn, adapt, and evolve—transforming it from a conceptual vision into the foundational unit of future human society.

3. AI EDGE: The Embodied Infrastructure of the Superintelligent Agent

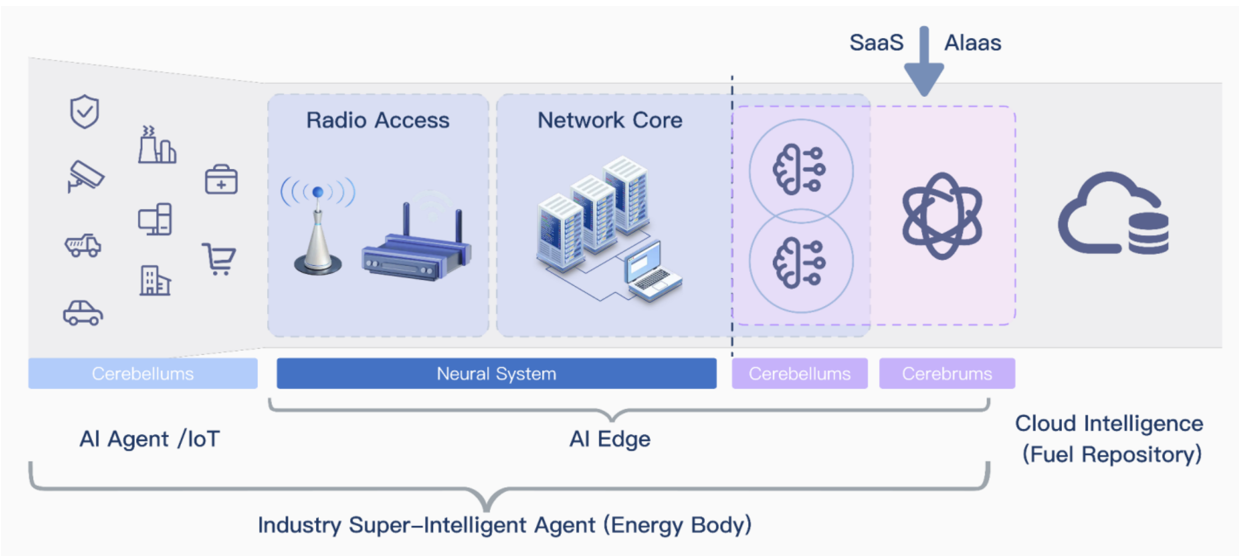

Figure 2 Super-Intelligent Agent Architecture: a) Central Control Unit (Brain): Responsible for global strategy, reasoning, and decision-making. It allocates tasks, synthesizes knowledge, and coordinates the entire system. b) Domain Modules (Cerebellums): Specialized sub-centers operating in vertical domains providing rapid responses through edge computing while feeding critical results back to the Brain, such as protein design, automated labs, video generation. c) Communication Network (Nervous System): A distributed infrastructure based on 5G/6G and edge computing, linking billions of "nerve endings" such as sensors, robots, and IoT devices. It ensures ultra-reliable, low-latency, and high-throughput connectivity. d) Physical Interfaces (Limbs): Embodied devices that perceive and act in the physical world. It ranges from robotic arms and surgical systems to drones and smart-city sensors. e) Driving Core (Energy Body): Distributed edge intelligence that executes perception–decision–action loops locally, enabling millisecond-level reflexes without waiting for central commands. f) Knowledge and Compute Base (Fuel Repository): Cloud and data centers that transform raw data into structured knowledge, train AI models (including the brain and downstreaming agents), and distill them into lightweight versions deployable at the edge.

AI Edge represents one of the most tangible and effective implementation paths toward the Superintelligent Agent. It transcends the traditional notion of wireless communication as mere data transmission, evolving into an intelligent substrate that fuses **communications, sensing, computation, intelligence, and control** into a coherent system tightly coupled with the physical world. Built upon mobile network infrastructures, AI Edge departs fundamentally from conventional 5G architectures. Rather than deploying terminals, base stations, and core networks separately, it co-locates radio access and network functions within *super-edge nodes*, creating a unified cyber-physical platform. Philosophically, AI Edge is task-oriented rather than connectivity-oriented: it delivers end-to-end services optimized for goals such as latency, safety, or yield, rather than generic throughput. This transformation embodies two dimensions—“more on communications” and “more than communications.” In the former, AI Edge achieves *hyper-converged processing* by integrating communication and sensing resources, forming the nervous system and limbs of the Superintelligent Agent. In the latter, it goes beyond transmission to exhibit *agentic self-learning and autonomous optimization*, continuously sensing environmental conditions, traffic dynamics, and user behaviors to self-adapt and evolve. Proximity to users and assets enables millisecond-level *perception–reasoning–execution* loops that power embodied applications like drones, robots, and autonomous vehicles—bridging the virtual and physical worlds in real time. Under tight constraints of latency, energy, and cost, AI Edge optimizes intelligence through multilevel co-design: compressed and distilled models at the algorithmic level, accelerated inference at the hardware level, and adaptive scheduling at the system level. Through distributed intelligence—via federated learning, knowledge distillation, split inference, and cross-scale model collaboration—it achieves the paradigm of a “global brain with agile reflexes”, where terminals handle reflexive decisions, edge nodes manage regional cognition, and the cloud orchestrates global understanding.

From Network Intelligence to Autonomous Outcomes

At scale, each AI Edge node functions as an autonomous agent with a perception–thinking–action loop, and the ensemble of such nodes forms a multi-agent system whose emergent behavior surpasses explicit programming. This architecture transforms network control from centralized management to decentralized governance: human engineers no longer act as direct controllers but as *game designers*, defining interaction protocols, resource-access policies, and multi-agent reinforcement learning reward functions to steer emergent behaviors toward safety, fairness, and energy efficiency.

Extending beyond application hosting, AI Edge also reconstructs the network itself through AI for Networks. By embedding deep learning into the radio access and core layers, the network becomes a *self-organizing organism* capable of adaptive optimization. AI-driven physical layers learn and predict channel states through intelligent estimation and Channel Knowledge Maps; RL-based resource management dynamically allocates spectrum and compute resources; and Deep RL-based network slicing maintains per-service QoS amid fluctuating demands. Through control–data plane unification, learning-based decisions co-reside with packet and signal processing on super-edge nodes, collapsing perception–decision–execution cycles. Intelligence is thus distributed across the body of the network, enhancing agility, scalability, and resilience—core promises of 6G. On the business and service layer, AI Edge introduces a new paradigm of autonomous outcome delivery: customers express intent, and the AI Edge Multi-agent system autonomously orchestrates communications, compute, and actuation to realize and sustain that goal. End-to-end intelligent services, on-demand edge provisioning, and a unified cloud–edge–terminal data plane dissolve traditional silos and enable continuous learning. Consequently, operators evolve from connectivity providers to solution brokers, marking the shift from selling bandwidth to delivering intelligent, self-adaptive outcomes—a defining milestone in the material evolution of the Superintelligent Agent.

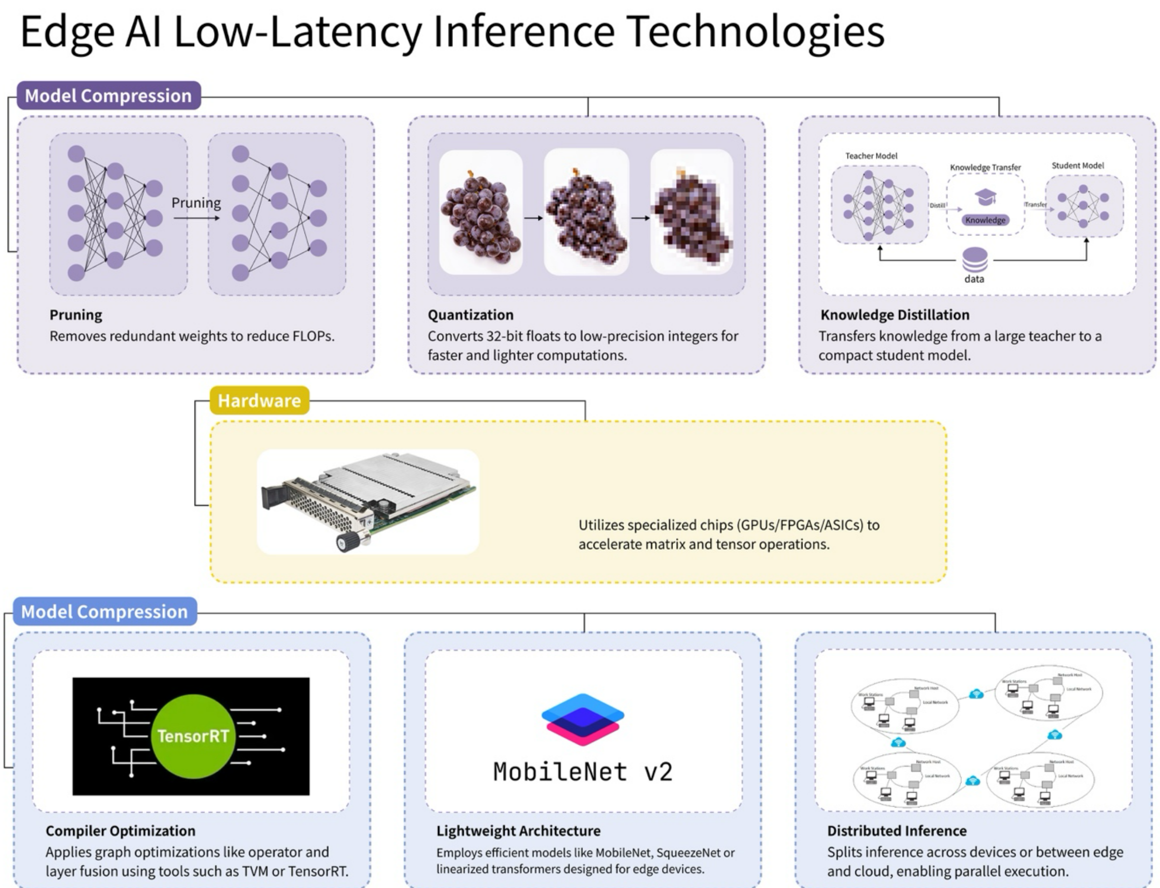

Figure 3 Overview of Edge AI Low-Latency Inference Technologies

4. Scientific Superintelligent Agent and New Paradigm of Discovery Automation

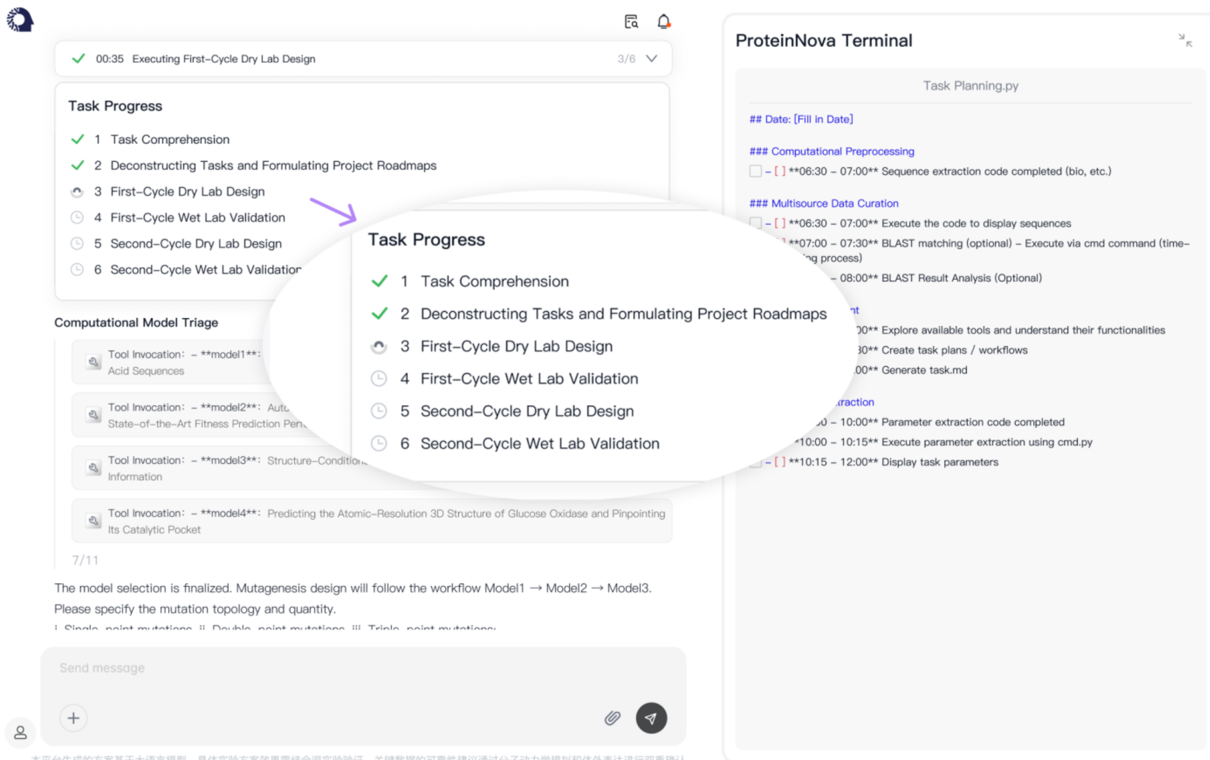

Figure 3 Superintelligent Agent for Innovative Drug R&D

The **Scientific Superintelligent Agent (SSA)** is the instantiation of the superintelligent agent within scientific research. Its mission is to accelerate discovery by surpassing human performance in hypothesis generation, experimental design, data analysis, and knowledge synthesis [3, 17-19, 64]. The disruptive leap arises from deep integration with automated wet labs, yielding a high-speed, iterative loop of *Hypothesize* → *Experiment* → *Analyze* → *Re-hypothesize*, which directly addresses long-standing bottlenecks in multivariable protocol design and optimization [20].

The Closed-Loop Paradigm for Pharmaceutical R&D

Innovative drug development suffers from high cost, long timelines, and low success rates, with traditional “trial-and-error” pipelines reaching diminishing returns. Recent advances show AI’s promise in early discovery—from molecular design to synthesis planning and property prediction [25-28]. As lab automation and AI mature, cross-disciplinary laboratories (chemistry, biochemistry, materials) are undergoing structural change: faster, more precise experiments unlock breakthroughs across health, energy, and electronics [30,54]. Against this backdrop, the SSA systematically upgrades pharma R&D by coupling its Brain with automated facilities:

- **Rapid Discovery.** The SSA integrates heterogeneous corpora—literature, omics, clinical records—to prioritize targets strongly associated with disease mechanisms. It then uses generative models (LLM/PLM) to propose large libraries of candidate molecules and performs in-silico triage (binding, ADMET, developability), compressing tasks that once required years into weeks or days [22].
- **Rapid Experimentation.** Protocols generated by the Brain are compiled to robotic workflows. Automated platforms conduct cell-based assays, HTS, and analytical readouts with closed-loop feedback. Outcomes are ingested in real time to adapt protocols and re-rank candidates. As programs advance, the Brain aids trial design—e.g., AI-assisted stratification and precision recruitment to reduce cost and risk.
- **Rapid Translation.** For promising assets, the SSA accelerates the path from "effective" to "usable": multi-omics integration clarifies mechanism of action (MoA); biomarker discovery supports patient selection and regulatory pathways, enabling precision medicine at scale.

From Tool Use to Domain-Native Intelligence: An Evolutionary Arc

Frontier frameworks reveal an evolution from *Tool User → Self-Evolving System → Domain-Native Multi-Agent*. The essence is a shift from syntactic orchestration to semantic, domain-grounded reasoning.

- **Stage 1 – Tool User (e.g.,** ChemCrow [32]).
  A general LLM orchestrates an expert-curated toolset via structured reasoning (*thought → action → observation*). Systems of this class can autonomously plan

multi-step procedures, invoke cheminformatics/synthesis tools, and execute on robotic platforms, with intelligence concentrated in planning and tool scheduling.

- **Stage 2 – Self-Evolving System** (e.g., Gemini Pro [65]).
  Beyond using tools, the agent creates new tools through meta-learning. A multi-agent architecture (e.g., Manager/Developer/Critic/Tool-Creator) maintains an evolving template library (successful reasoning patterns) and a tool ocean that expands by discovering and integrating domain tools on demand—thus adapting capabilities to novel tasks and datasets [50].
- **Stage 3 –** Domain-Native Multi-Agent Framework (e.g., ProteinNova).
  The "brain" is no longer a generic LLM but a multimodal, domain-native foundation model with intrinsic understanding of protein biophysics. For example, the TourSynbio-7B model is trained on mixed corpora of text and protein sequences (PLM + NLP), enabling the system to compose knowledge-driven workflows rather than merely decompose tasks. In wet-lab validation, such frameworks have reported significant gains (e.g., increased P450 19-hydroxylation selectivity for steroids; multi-fold improvements in reductase catalytic efficiency) [51-52]. While specific outcomes vary by dataset and assay, the direction is clear: SSA is migrating from assistant to coworker scientist.

Why SSA Matters Beyond Pharma

Although pharmaceuticals provide a vivid testbed, the SSA paradigm generalizes to materials discovery, synthetic biology, energy systems, and environmental science. The common pattern is a DBTL loop driven by domain-native models and automated experimentation**:** Design–Build–Test–Learn compresses program cycles and amplifies scientific throughput. As domain models become stronger (physics-informed, multimodal, interventional), SSAs will be capable of proposing hypotheses, designing critical experiments, and refuting/confirming them at machine timescales—re-shaping epistemic practice.

Governance, Credit, and Epistemic Shifts

SSA forces a reconsideration of authorship, accountability, and reproducibility. If algorithms propose hypotheses, design experiments, and perform analysis, how are credit and responsibility assigned among human teams, institutions, and autonomous agents? Moreover, alignment and safety become scientific concerns: SSAs must respect experimental constraints, biosafety/regulatory guardrails, and data governance. We argue that lab accreditation and publication practices should evolve to recognize agent-in-the-loop workflows, including provenance tracking, policy-constrained planning, and audit trails for autonomous actions.

5. The Industrial Superintelligent Agent: The Revolutionary Force for Reshaping Future Industry

Industrial and logistics robotics, empowered by 5G/6G connectivity, are gaining freedom and autonomy in complex environments [41-43]. Yet hardware advances alone are insufficient. The true transformation arises when AI integrates across the entire value chain, reshaping production from design to delivery. The Industrial Superintelligent Agent (ISA) is the embodiment of this vision: a system with perception, cognition, decision-making, and execution that exceeds human expertise, spanning macroscopic supply chain orchestration to microscopic defect detection. Its purpose is not to automate tasks but to liberate humanity from repetitive toil, creating space for creativity and strategy.

Core Value: Productivity Liberation and Systemic Efficiency

The ISA's value manifests on two levels:

- **Level 1 – Achieving Extreme Energy Efficiency.** Facing resource scarcity and personalized demand, ISAs enable forward-looking design beyond human heuristics, making products adaptable across scenarios. By coordinating device–edge–cloud intelligence and balancing loads in real time, they minimize resource use while maximizing throughput, delivering unprecedented efficiency gains.

- **Level 2 – Building Collaborative Production Paradigms.** ISAs establish ecosystems of interconnected agents. AI-driven “designer agents” accelerate product innovation; ubiquitous sensors and robots execute operations; and robust 5G/6G + edge infrastructures sustain low-latency, high-reliability coordination. This new paradigm of multi-agent collaboration redefines industrial relations.

Bottlenecks of Traditional Industry

Despite centuries of progress, traditional industrial models face mounting challenges:

- **Labor shortages and rising costs.** High-intensity, hazardous jobs are increasingly unattractive, straining labor supply.
- **Rigid, low-flexibility production.** Legacy lines, optimized for mass production, cannot efficiently handle small-batch, personalized demand.
- **Fragile supply chains.** Global complexity makes systems vulnerable to geopolitical or natural disruptions.
- **Energy and carbon constraints.** Heavy industries face regulatory and cost pressures to achieve carbon neutrality.
- **Inconsistent quality control.** Reliance on human inspection yields inefficiencies, errors, and lack of precision for advanced manufacturing.

These pressures make **intelligent transformation a necessity, not an option.**

Empowering Interconnected Agents

Interconnection is the foundation of ISA-driven transformation:

- **AI EDGE platforms.** Integrated super-nodes unify communication, sensing, and computation, breaking silos between proprietary devices and protocols.
- **Real-time data flow.** URLLC + mMTC capabilities of 5G/6G, coupled with edge computing, ensure millisecond-level responsiveness for mission-critical control [35-36, 38].

- **Semantic interoperability.** ISAs harmonize "dialects" of heterogeneous agents—AGVs, robotic arms, cameras—into a common task language**,** enabling semantic understanding and coordination.

Achieving Collaborative Production

On this foundation, ISAs orchestrate:

- **Intelligent scheduling.** Like operations-research experts, ISAs process live data on orders, inventories, equipment, and energy to generate globally optimal production plans in seconds.
- **Robot swarm coordination.** In complex assembly, ISAs dynamically assign roles, plan paths, and prevent collisions across mixed robot fleets.
- **Digital twins.** Virtual replicas of factories and supply chains allow simulation, diagnosis, and optimization, reducing trial-and-error costs.
- **Predictive maintenance.** AI models anticipate component failures weeks ahead, scheduling downtime proactively to maximize overall equipment effectiveness.
- **End-to-end quality control.** AI vision and sensor fusion enable 100% real-time inspection, shifting quality management from reactive inspection to proactive prevention**.**

Toward Dark and Smart Factories

The long-term trajectory of ISA development is the realization of dark factories and *smart factories*—autonomous environments where orders, procurement, scheduling, production, logistics, and energy management are all governed by ISAs. Robots and automation systems execute physical work, while humans transition to supervisors, creators, and strategists, focusing on innovation and governance.

In this future, industrial superintelligence is not a supplement to human labor but a fundamental redesign of industrial organization, reconciling efficiency with sustainability and freeing human potential for higher-order pursuits.

The Superintelligent Agent demonstrates transformative potential across manufacturing, finance, healthcare, logistics, and energy: it streamlines production, reduces financial and operational risks, enhances diagnostic accuracy and personalized treatment, accelerates drug and technology discovery, optimizes supply chains, and drives sustainable energy development—collectively advancing industrial intelligence and societal benefit.

6. Conclusion

This paper has outlined the vision, architecture, and implications of the Superintelligent Agent as the prospective fundamental unit of scientific and industrial life. In theory, the SIA is a system that integrates perception, cognition, decision-making, and execution, surpassing human cognitive limits. In practice, its two instantiations point to complementary futures: the Scientific Superintelligent Agent, which—through integration with automated laboratories—promises orders-of-magnitude acceleration in discovery; and the Industrial Superintelligent Agent, which—by restructuring production processes—seeks to liberate humanity from repetitive labor and stimulate creativity.

We deconstruct the AI Edge architecture as the enabling substrate for this vision. Several paradigm shifts emerge:

- From data-driven to physics-infused. Foundation models must go beyond pattern recognition to become digital twins that capture the physical world.
- From centralized control to collective intelligence. The AI Edge network functions as a multi-agent ecosystem, governed not by imperative commands but by interaction rules and incentive mechanisms, making emergent governance the central challenge.
- From network-as-carrier to network-native intelligence. With AI4NET, intelligence is embedded into the infrastructure itself, unifying control and data planes and transforming the network into a distributed, self-optimizing organism.

- From selling resources to delivering outcomes. The business model shifts from commoditized compute/connectivity to intent-driven services, elevating providers into solution brokers of autonomous results**.**

The implementation pathway is challenging yet concrete: begin with AI for Science, using domain-specific problems to mature hardware–software integration of AI Edge with automated experimentation; then generalize to AI for Industry, enabling robotic collaboration and adaptive design. Together, these domains can progressively fulfill the grand mission of liberating human productivity.

At the same time, the advent of superintelligence carries profound responsibility. Questions of ethics, alignment, accountability, and societal impact must remain central. Without prudent governance, the very systems designed to amplify human creativity could undermine it. With careful stewardship, the Superintelligent Agent holds the promise of inaugurating a new epoch of scientific progress, industrial renewal, and human flourishing.

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